Human Language Technology Research Institute

Unveiling Hidden Intentions

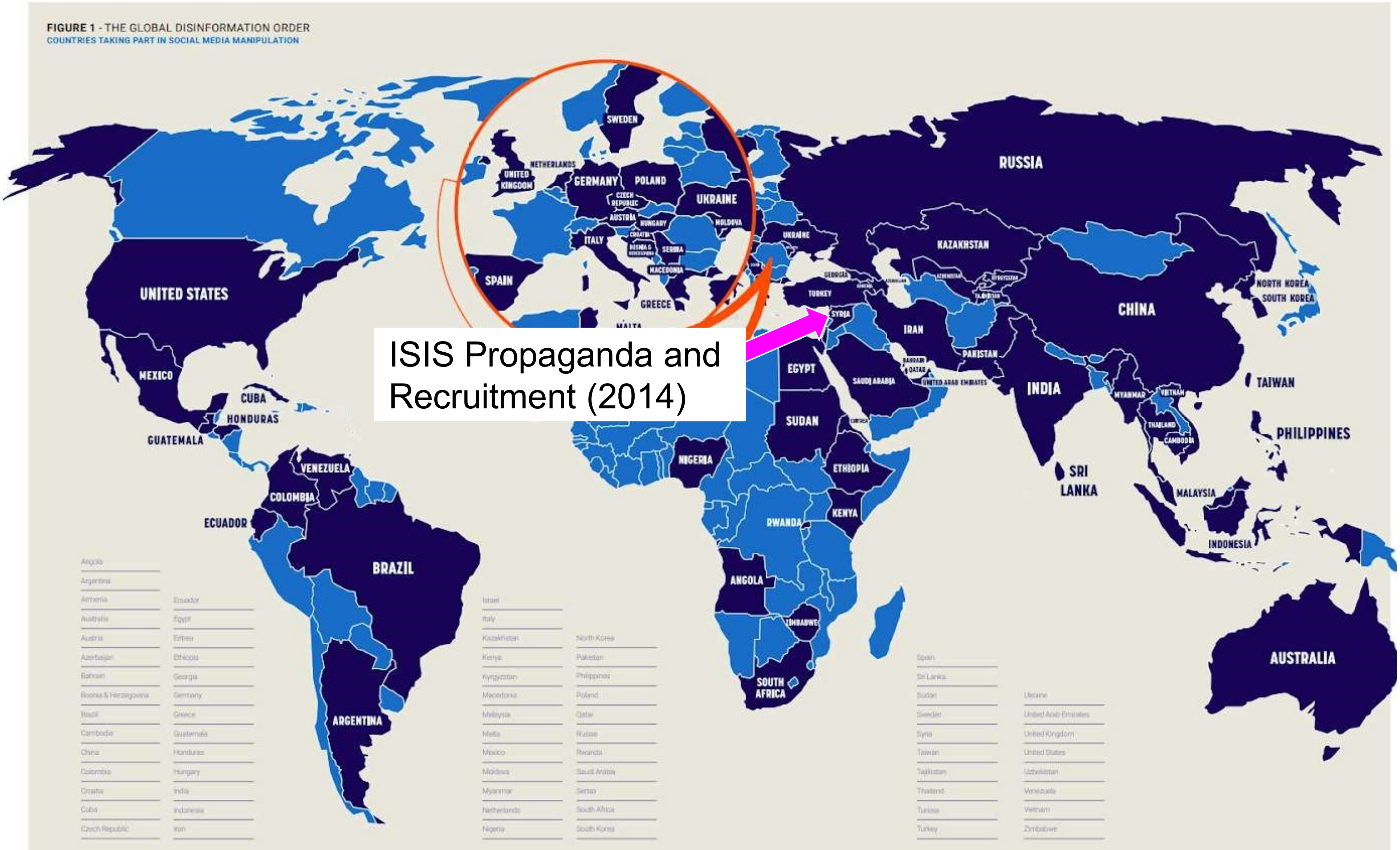
Gerardo Ocampo Diaz and Vincent Ng

University of Texas at Dallas

AAAI 2020

Social Media Manipulation

Social Media Manipulation



Social Media Manipulation

Social Media Manipulation

Pro-government content on Twitter and Facebook (2019 Hong Kong Protests)

What are we doing about this?

- Automatic identification of bots and fake accounts
- Detection of fake news and Deepfake videos

Are we doing enough?

- No!
- Content needs not be fake to be malicious

Framing

- The way the media **cherrypicks** and **presents** information to the audience with the goal of influencing how they process it

Framing

- The way media **cherrypicks** and **presents** information to the audience with the goal of influencing how they process it

- E.g., the 2019 Hong Kong protests
  - A report focusing on the police's use of weapons and the number of injuries as a result
    - Negative sentiment on the police

Framing

- The way media **cherrypicks** and **presents** information to the audience with the goal of influencing how they process it

- E.g., the 2019 Hong Kong protests
  - A report focusing on the police's use of weapons and the number of injuries as a result
    - Negative sentiment on the police
  - A report focusing on how violent the protesters were and how economy has suffered as a result
    - Negative sentiment on the protesters

Proposed Task:
Hidden Message and Intent Identification

- Given an input (e.g., text, video, image), generate a short description of the hidden message conveyed by it and the author's intention, if any

Surface vs. Hidden Message/Intention

- E.g., the 2019 Hong Kong protests
  - **Surface message/intention**
    - Report on the protest in Hong Kong with no particular messages other than the facts reported
  - **Hidden message/intention**
    - Depends on how the protest was framed
      - Message: the protesters are criminals
      - Intent: discredit the protesters, support the government

Surface vs. Hidden Message/Intention

- E.g., the 2019 Hong Kong protests
  - **Surface message/intention**
    - Report on the protest in Hong Kong with no particular messages other than the facts reported
  - **Hidden message/intention**
    - Depends on how the protest was framed
      - Message: the protesters are criminals
      - Intent: discredit the protesters, support the government

Alleviating Political Manipulation

- Reveal the hidden intent of the author, if any, to the reader

Broader Implications for AI

- Significant advances in machine perception in recent years
  - AI systems can now “see” and “read”
  - But… they still cannot read between the lines

- Automatic understanding of hidden intents and messages requires a deeper level of understanding
  - Enables a machine to get a step closer to human perception

Broader Implications for AI

- Significant advances in machine perception in recent years
  - AI systems can now “see” and “read”
  - But… they still cannot read between the lines

- Automatic understanding of hidden intents and messages requires a deeper level of understanding
  - Enables a machine to get a step closer to human perception

Who knows the author's real intent?

- No one other than the author can say for sure what the author's real hidden intention and message are

- Realistically… the task involves identifying his hidden intention and message **as perceived by the audience**

Task Input

- Text
- Image
- Video

Task Input

- Text
- Image
- Video

An Example

President Fraser Pushes for a New Trade Law

In the wake of the last economic recession, which started just a few months after President Fraser's energetic reform bill was passed into law,

the president has announced a new bill that constitutes a complete overhaul of trade law which he claims will help strengthen the economy;

An Example

President Fraser Pushes for a New Trade Law

In the wake of the last economic recession, which started just a few months after President Fraser's energetic reform bill was passed into law,

the president has announced a new bill that constitutes a complete overhaul of trade law which he claims will help strengthen the economy;

Fraser’s bill caused the last economic recession

An Example (Cont')

But Republicans aren't having any of it.

Republicans successfully backed a series of bills that helped keep the economy afloat during the recession,

and **they claim that the new reform will only make matters worse.**

An Example (Cont’)

But Republicans aren't having any of it.

Republicans successfully backed a series of bills that helped keep the economy afloat during the recession,

and **they claim that the new reform will only make matters worse.**

Republicans are trustworthy

The hidden message

The new reform bill should not be passed into law.

In the wake of the last recession, which started a few months after President Fraser's reform bill…

Fraser's reform caused the last recession

In the wake of the last recession, which started a few months after President Fraser's reform bill…

Fraser's reform caused the last recession

The president has announced a new bill that results in an overhaul of trade law…

In the wake of the last recession, which started a few months after President Fraser's reform bill…

Fraser's reform caused the last recession

The president has announced a new bill that results in an overhaul of trade law…

In the wake of the last recession, which started a few months after President Fraser's reform bill…

Republicans successfully backed a series of bills that helped the economy…

Fraser's reform caused the last recession

The president has announced a new bill that results in an overhaul of trade law…

Republicans are trustworthy

In the wake of the last recession, which started a few months after President Fraser's reform bill…

Republicans successfully backed a series of bills that helped the economy…

Fraser's reform caused the last recession

The president has announced a new bill that results in an overhaul of trade law…

Republicans are trustworthy

They claim the new reform will make matters worse

In the wake of the last recession, which started a few months after President Fraser's reform bill…

Republicans successfully backed a series of bills that helped the economy…

…

Fraser’s reform caused the last recession

In the wake of the last recession, which started a few months after President Fraser’s reform bill…

The president has announced a new bill that results in an overhaul of trade law…

Republicans are trustworthy

Republicans successfully backed a series of bills that helped the economy…

They claim the new reform will make matters worse

The new bill should not be passed into law.

Fraser's reform caused the last recession

In the wake of the last recession, which started a few months after President Fraser's reform bill…

The president has announced a new bill that results in an overhaul of trade law…

Republicans are trustworthy

Republicans successfully backed a series of bills that helped the economy…

They claim the new reform will make matters worse

. . .

The new reform bill should not be passed into law.

. . .

Fraser's reform caused the last recession

The president has announced a new bill that results in an overhaul of trade law…

Republicans are trustworthy

They claim the new reform will make matters worse

In the wake of the last recession, which started a few months after President Fraser's reform bill…

Republicans successfully backed a series of bills that helped the economy…

The new reform bill should not be passed into law.

. . .

Fraser's reform caused the last recession

The president has announced a new bill that results in an overhaul of trade law…

Republicans are trustworthy

They claim the new reform will make matters worse

In the wake of the last recession, which started a few months after President Fraser's reform bill…

Republicans successfully backed a series of bills that helped the economy…

Common Tactics

- The author implicitly argues for his/her (hidden) message, often without sounding argumentative, via:
  - framing
  - exploiting logical fallacies

Related Work: Textual Entailment

- Given a pair of statements (a text and a hypothesis), determine whether the hypothesis follows from the text

Text: If you help the needy, God will reward you.

Hypothesis: Giving money to a poor man has good consequences.

Related Work: Textual Entailment

- Given a pair of statements (a text and a hypothesis), determine whether the hypothesis follows from the text

Text: If you help the needy, God will reward you.

Hypothesis: Giving money to a poor man has good consequences.

- Classification rather than generation task
- Useful for constructing our argument tree
  - A entails B → B could be A's parent

Related Work: Causal Relation Extraction

- Extract text segments that are causally related in a discourse

The Dow Jones Index dropped by more than 10% today. Investors are in fear of an economic recession.

Related Work: Causal Relation Extraction

- Extract text segments that are causally related in a discourse

The Dow Jones Index dropped by more than 10% today. Investors are in fear of an economic recession.

- Extraction rather than generation task
- Useful for constructing argument trees
  - A causes B → B could be A's parent

Related Work: Argument Mining

- Build an argument tree from argumentative text
- Differs from hidden intent and message identification
  - Input is argumentative text
  - Tree nodes can contain only text segments from the input
    - Nothing hidden
      - Extraction rather than generation task

Related Work: Entity-level Stance Detection

- Given a text, determine the author's stance (for, against) on the entity mentioned in the text

President Gee is an awful person.

- (President Gee, against)

Related Work: Entity-level Stance Detection

- Given a text, determine the author's stance (for, against) on the entity mentioned in the text

President Gee is an awful person.

- (President Gee, against)

- Useful for understanding the author's intent
  - What sentiment is s/he trying to express towards which entity?
- Useful for constructing an argument tree
  - Assemble evidences consistent with his/her stance on an entity

How might we approach this task?

- **Supervised**
  - Learn from a training set where each instance corresponds to a text paired with the hidden message (and intention)

How might we approach this task?

- **Supervised**
  - Learn from a training set where each instance corresponds to a text paired with the hidden message (and intention)
- How to create this training set?
  - **Crowdsourcing** may be needed for generation tasks
    - Annotation quality might be a concern
      - space of hidden messages is huge: can people agree?
      - overly general messages (X is good, X is bad, …)
      - multiple correct messages
      - same message written in different ways

How might we approach this task?

- **Supervised**
  - Learn from a training set where each instance corresponds to a text paired with the hidden message (and intention)
- How to create this training set?
  - **Crowdsourcing** may be needed for generation tasks
    - Annotation quality might be a concern
      - space of hidden messages is huge: can people agree?
      - overly general messages (X is good, X is bad, …)
      - multiple correct messages
      - same message written in different ways
- May still be a challenging learning task

Attempt 2: Mimic Human Reasoning

- Build an argument tree from the bottom up
- May need an annotated training set where each instance corresponds to a text and the associated argument tree
  - Even less agreement among human annotators?

Attempt 2: Mimic Human Reasoning

- Build an argument tree from the bottom up
- May need an annotated training set where each instance corresponds to a text and the associated argument tree
  - Even less agreement among human annotators?
- **Learning task**
  - How to combine evidences to derive a (possibly hidden) conclusion that results in a persuasive argument?
  - Related tasks (textual entailment, argument mining) may help
    - But each of these tasks is challenging
    - Errors made earlier may propagate up the tree

Attempt 3: Only build the top 2 levels of tree

- Given an input text,
  1) **Topic Modeling**
     - infer the main entity/topic
  2) **Stance Detection**
     - determine the author’s stance towards the main topic/entity
  3) **Argument Mining**
     - collect text segments that support the author’s stance towards the main entity
  4) **Text Generation**
     - generate the hidden message from these text segments

Attempt 3: Only build the top 2 levels of tree

- Given an input text,
  1) **Topic Modeling**
     - infer the main entity/topic
  2) **Stance Detection**
     - determine the author's stance towards the main topic/entity
  3) **Argument Mining**
     - collect text segments that support the author's stance towards the main entity
  4) **Text Generation**
     - generate the hidden message from these text segments

- Training set: annotation of top two levels of the tree only
- But… learning to build the top 2 levels is still challenging

Text Generation

- Remains a challenging task
- May need **background knowledge**

Text Generation

- Remains a challenging task

- May need **background knowledge**
  - What was the significant event taking place when the input was composed?
    - Context that may be missing from social media text
  - How are the entities and events related to each other?
    - E.g., in the 2019 Hong Kong protests
      - supporting the police → pro-government

Text Generation

- Remains a challenging task
- May need **background knowledge**
  - Knowledge acquisition is a challenging problem

Memes and Images: Example

- **Hidden message**
  - A vote for Hillary is a vote for Satan
- **Intention**
  - Encourage people to vote for Trump
- **Premises**
  - Satan and Jesus are enemies
  - Satan is Bad, Jesus is Good
  - Hillary Clinton and Donald Trump are rivals
- **Message**
  - Satan and Hillary are friends/partners

Memes and Images: Example

- **Hidden message**
  - A vote for Hillary is a vote for Satan
- **Intention**
  - Encourage people to vote for Trump
- **Premises**
  - Satan and Jesus are enemies
  - Satan is Bad, Jesus is Good
  - Hillary Clinton and Donald Trump are rivals
- **Message**
  - Satan and Hillary are friends/partners

Memes and Images: Example

- **Hidden message**
  - Millenials are considering socialism over freedom and democracy, which they should not
- **Intention**
  - Criticize millennials / Establish socialism as incompatible with freedom and democracy
- **Premises**
  - The girlfriend represents “correct” or just beliefs
  - The other girl represents an alternative set of beliefs
  - The girlfriend should be preferred over other girls.

Summary

- Proposed Hidden Intention and Message Identification task

- More complex than other well-known challenging reasoning tasks such as the Winograd Schema Challenge
  - Each type of input (text, image, ..) has its own set of challenges

- Could enable a machine to “read between the lines” and hence move one step closer to human perception